

Varun Raghuraj Rana

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Machine Learning Engineer | Quantitative Analytics & Computer Vision | BU MS Data Science '27

EDUCATION

Boston University, Faculty of Computing & Data Sciences
Master of Science in Data Science

Boston, MA
Expected Dec 2027

GPA: 4.00/4.00 | **Relevant Coursework:** Graduate Algorithms, Deep Learning, Mathematical Statistics, Stochastic Processes, Cryptography for Data Science, AI Ethics, Stochastic Methods for Algorithms

Parul University

Bachelor of Technology in Chemical Engineering; Diploma in Neural Networks & Deep Learning

Vadodara, India
Aug 2021 – May 2025

SKILLS

- **Programming & Frameworks:** Python (NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, Matplotlib, Seaborn), C++, JavaScript, TypeScript, SQL
- **ML & Computer Vision Tools:** YOLO, OpenCV, CNNs, RNNs, GANs, Transformers, Hyperparameter Tuning, Model Optimization, Class Imbalance Handling
- **Data & Deployment:** MongoDB, Azure Blob Storage, Docker, FastAPI, Flask, Gradio, REST APIs, Next.js, WebSockets, Git/GitHub
- **Quantitative & Analytics:** Time-Series Analysis, Statistical Modeling, Backtesting, Bayesian Optimization (Optuna), EDA, Tableau, Power BI

PROFESSIONAL EXPERIENCE

AI/ML Intern; SETV Global

Vadodara, India
Aug 2025 – Jan 2026

- Built and trained computer vision models for gastrointestinal, oncology, neurology, and musculoskeletal diagnostics using YOLO and ConvNeXt, achieving >88% evaluation accuracy across clinical datasets.
- Created a standardized end-to-end ML experimentation pipeline (data mining, annotation, EDA, class balancing, hyperparameter tuning, training/testing) reducing model iteration cycles by ~25%.
- Developed deployable medical AI tools using Flask and Gradio enabling real-time inference, automated PDF diagnostic reports, and historical scan comparison.
- Implemented scalable imaging data pipelines using Azure Blob Storage and MongoDB for structured storage and metadata tracking of medical datasets.

Operations Intern; Adani Hazira Port Limited (AHPL)

Hazira, India
Jun 2024 – Jul 2024

- Processed 200+ cargo documentation records monthly while maintaining compliance with industrial logistics standards.
- Monitored tanker loading/unloading using SCADA systems and assisted in vessel-to-tank operations handling 500M+ liters of liquid cargo.

PROJECTS

Algorithmic Trading Strategy Engine & FinSightAI Dashboard

- Designed an end-to-end multi-asset algorithmic trading bot featuring 10 production-ready strategies and 16 advanced quant modules, integrated with FinSight AI, a full-stack dashboard for real-time equity tracking and portfolio analytics.
- Achieved a 4.66 Sharpe Ratio and 80% win rate in optimized 6-month backtests; attained an "A+" stability grade by passing 46/46 extreme market stress tests (flash crashes, high-frequency loads).
- Engineered a custom C++ mathematical engine delivering a 188× execution speedup; deployed a live paper-trading environment with dynamic risk management (circuit breakers, correlation guards, Kelly-inspired position sizing).

Gastrointestinal Disease Detection Model

- Developed a YOLO-based object detection model for stomach ulcers and polyps in endoscopic images, achieving mAP of 0.90 across classes and subclasses with augmentation-driven robustness and clinical-grade interpretability.

Medical X-Ray Generation & Report System

- Built a hybrid GAN + NLP pipeline to generate synthetic X-ray images (high SSIM) and automated diagnostic reports (BLEU 0.87), combining adversarial image synthesis with transformer-based text generation.

Green-Synthesized Nano Zero-Valent Iron (nZVI) for Sustainable Environmental Remediation

- Investigated plant-mediated nZVI synthesis; achieved 92.3% dye removal. Characterized via SEM, EDS, XRD, FTIR; kinetic analysis fit pseudo-second-order model ($R^2 = 0.9994$). Poster presented at ICCMSE 2023.

LEADERSHIP & ACHIEVEMENTS

NCC C-Certificate Holder, Ministry of Defence, GoI — 3 years military training; represented university at Special National Integration Camp (Jorhat) & All India Ship Attachment Camp (Mumbai Dockyard).

3rd Dan Black Belt, Iran Karate-Do Federation — 2× Gold, 3× Silver, 3× Bronze at National & State Championships; represented Parul University at AIU Championship 2022.